

Management of Spinal Neurogenic Bladder & Bowel

UCLH Guideline

Trust Wide

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1.0 Summary

- 1.1 This guidance sets out the agreed management pathways for patients who have spinal cord neurogenic bladder and / or bowel; it has been developed based on best clinical evidence and includes a specific care bundle to support the nursing care of these complex patients. It aims to provide clear guidance to aid staff in the timely and appropriate assessment, management and treatment of this patient cohort.
- 1.2 It emphasises the importance of appropriate bowel and bladder care in this group of patients. In addition, it gives guidance in the care of patients with an established spinal cord injury above the Level of T6, who have Autonomic Dysreflexia (AD).
- 1.3 The guidance applies to all nursing staff involved in the care of spinal patients and in particular to:-

NHNN – neurosurgical and neurology wards, theatres, recovery and critical care.

2.0 Introduction

- 2.1 The care for patients with spinal cord injury is complex and requires specialist knowledge and skills. Patients with spinal cord injury (SCI) will have a neurogenic bladder and bowel. This must be both recognised and managed appropriately as soon as SCI is diagnosed or suspected
- 2.2 Over a thousand spinal operations are performed at NHNN annually; over 100 of these will have some form of SCI.
- 2.3 Routinely, patients with spinal cord problems, e.g. patients with metastatic spinal cord compression will be admitted elsewhere within the trust and will also have bladder and bowel problems secondary to their damaged spinal cord.
- 2.4 There is a complex spine team at NHNN. The team comprises of complex spinal neurosurgeons and specialist nurses.
<https://www.uclh.nhs.uk/OurServices/ServiceAZ/Neuro/BTU/Pages/Consultants.aspx>
- 2.5 In addition, non-complex spinal neurosurgeons may perform emergency (Particularly relevant for Cauda Equina surgery) spinal surgery.
- 2.6 The spinal nursing team recognised the difficulties that non-specialist nurses and medical staff had in identifying, assessing and

planning treatment for these patients. Following a review of available evidence developed this guideline and the bladder and bowel pathways (See appendices 1 and 2) to provide optimal, evidenced based care to SCI patients.

3.0 Objectives

- 3.1 This guidance sets treatment pathways (appendices 1 and 2) for the immediate management of bowel and bladder function in spinal cord injured patients.
- 3.2 These pathways support the timely, appropriate, assessment, management and treatment of patients presenting with a SCI at NHNN, ensuring optimal bowel and bladder care.

4.0 Scope

This policy applies to the University College London Hospitals and Neurosurgery and to all clinical staff involved in the care of spinal patients and in particular to:-

- Nursing Staff
- Surgical, medical and anesthetic staff
- recovery, critical care, neurosurgical, and neurology wards, theatres, radiology, anaesthetics, therapies and ward pharmacists.
- Patients with a new diagnosis SCI.
- Patients with an established SCI - who are at risk of Autonomic Dysreflexia (AD)

5.0 Definitions

- 5.1 Cauda equina Syndrome – a spinal cord injury affecting the nerves of the cauda equina only
- 5.2 Neurological deficit – power or sensation loss (either complete or partial)
- 5.3 Spinal cord injury – interruption of the normal spinal cord function (either complete or partial)
- 5.4 Autonomic Dysreflexia – Malignant hypertension, secondary to a noxious stimuli, below the level of SCI (usually neurological level T6 and above)
- 5.5 Neurogenic bowel – dysfunction of the bowel due to the loss of normal sensory and/or motor control as a result of SCI
- 5.6 Neurogenic bladder - dysfunction of the bladder due to the loss of normal sensory and/or motor control as a result of SCI
- 5.7 Areflexic – complete cessation of reflex activity due to damage at or below the level of the bottom of the spinal cord (conus medallaris)
- 5.8 Reflexic – Intact reflexes below the level of the injury, as the damage to the cord is above the level of the conus medallaris
- 5.9 Manual evacuation - Digital removal faeces (DRF)

6.0 Acronym List

Acronyms used in guideline and appendices

- 6.1 AD - Autonomic Dysreflexia
- 6.2 ASIA - American Spinal Injuries Association neurological score
- 6.3 DRE – Digital Rectal Examination
- 6.4 DRF - Digital removal faeces or manual evacuation of stool
- 6.5 DSD - Detrusor Sphincter Dysynergia
- 6.6 ISC – Intermittent Self Catheterization
- 6.7 IC – Intermittent Catheterization
- 6.8 NDO - Neurogenic Detrusor Overactivity
- 6.9 NLI – Neurological Level of Injury
- 6.10 SCI - spinal cord injury
- 6.11 SPC – Suprapubic catheter
- 6.12 TWOC – Trial without catheter

7.0 Duties & Responsibilities

- 7.1 All nursing, medical and pharmacy staff who are involved in the care of spinal patients should be familiar with this guideline
- 7.2 The Consultant, and the nurse in charge of an individual patient, are responsible for ensuring that the patient is on the correct pathway and receiving the correct care.
- 7.3 The education team are responsible for disseminating the information contained within this guideline and providing the appropriate levels of training to the staff involved in the care of spinal patients.
- 7.4 The neurosurgical team and specialist spinal nursing team are responsible for providing expert advice and support in the care of spinal patients.
- 7.5 Clinical guideline and other governance approval committees are responsible for reviewing the documentation and providing clinical governance support.

8.0 Development & Evidence Base

- 8.1 This guideline has been developed to provide exemplary, evidenced based bladder and bowel care to SCI patients.
- 8.2 The guideline has been adapted from the best practice guidance from the Multidisciplinary Association of Spinal Cord Injury Professional (MASCIP) 2012, the NICE guideline 'Improving outcomes for people with brain and other central nervous system tumours' ([CSG10] 2006, reviewed 2017).
- 8.3 The specific bladder and bowel care guidance is supplemented by the NICE guideline 'Metastatic Spinal Cord Compression' (2008, updated 2017), that gives recommendations on the optimal care of patients with MSCC. This guidance is supported by the NICE guideline 'Spinal Injury Management and Initial Assessment' [NG41] (2016) that gives clear recommendations on the optimum management of spinal pathways for patients with spinal injuries.
- 8.4 Additionally, this guidance incorporates the recommendations made by Patients safety alert 'Resources to support safer bowel care for patients at risk of autonomic dysreflexia' (2018) to improve the care of patients with a SCI and AD (see appendix 7)

9.0 Consultation

- 9.1 Development of these local care pathways and associated care guidance, has been led by the Spinal Nursing team at NHNN.
- 9.2 Approval has been obtained from appropriate committees and groups: Queen Square Clinical Governance Committee,

10.0 Bowel and Bladder care in Patients with a Spinal Cord Injury

This guidance has been designed so that a patient with a SCI is clinically assessed and given the appropriate bowel and bladder care in a prompt, safe and timely manner.

- Patients with SCI will have neurogenic bladder and bowel dysfunction caused by the damage to the spinal cord. Both bladder and bowel dysfunction can be broadly divided into two categories, that is reflexic and areflexic.

- The type of bowel and bladder regime needed will depend on the level of spinal cord injury. The level of the injury is determined by a physical examination by the medical staff, known as the American Spinal Injuries Association (ASIA) score (see appendix 6).
- Once patients have had their ASIA completed, a nursing and medical plan for their bladder and bowel care should commence immediately.
- The level of injury will determine the type of bladder and bowel care that patient will require. Nursing and medical staff are to consult the appropriate pathway and start appropriate management regimes:

See Appendix 1 for bladder pathway. A comprehensive description of the nursing and medical interventions required for the pathway, alongside the rationale and evidence for these actions are found in appendix 3. Please refer to trust guidance on fluid balance documentation.

See Appendix 2 for bowel pathway. A comprehensive description of the nursing and medical interventions required for the pathway, alongside the rationale and evidence for these actions are found in appendix 4. This documentation includes guidance on the drugs that should be prescribed initially and this regimen should be reviewed after 72 hours.

- Specialist spinal nursing advice will be required if the level of injury is directly at the conus medullaris. These, rare, patients can present with a mixed bladder and bowel pattern; including elements of both reflexive and areflexive function. With these patients, the areflexive pathway should be followed until specialist spinal nursing advice is obtained.
- Patients that are admitted to the hospital with an established spinal cord injury will have a bladder and bowel regime in place. As highlighted in the recent National Patients' Safety Alert (see appendix 7) continuation of their bladder and bowel regime is extremely important as mismanagement can result in the life threatening condition - Autonomic Dysreflexia.

- AD is a potentially dangerous complication for patients with a spinal cord injury (SCI) at T6 or above. In AD, an individual's blood pressure may rise to dangerous levels and if not treated can lead to stroke and possibly death. AD usually occurs when there is a noxious (irritating) stimulus below the level of the injury. Bladder distention, blocked catheters and urinary tract infections are frequent causes of AD. Constipation, poorly managed bowel care and digital removal of faeces can be causes of AD. Symptoms include headache, tachycardia or bradycardia, facial flush, perspiration, and a stuffy nose (United Spinal Association, 2015).
- It is imperative that patients with an established cord injury are able to continue their normal bladder and bowel management, including trans anal irrigation. Patients may wish to have their own personal carers perform their bladder/bowel regime while inpatients, including manual evacuation. If patients chose to have their own careers this should be facilitated by the ward and be part of the patients care plan.

The care of patients with spinal neurogenic SCI is supported by the following documents and resources:

1. Acute spinal cord injury (ASCI) bladder management Flow diagram that simplifies the specific care pathways that the patient needs for optimal treatment following a diagnosis of neurogenic bladder(appendix 1)
2. ASCI bowel management Flow diagram that simplifies the specific care pathways that the patient needs for optimal treatment following a diagnosis of neurogenic bowel (appendix 2)
3. A comprehensive bundle that outlines the best evidence based care - 'Bladder care in acute spinal cord injury' (appendix 3)
4. A comprehensive bundle that outlines the best evidence based care - 'Bowel care in acute spinal cord injury' (appendix 4)
5. Bristol stool chart (appendix 5)
6. ASIA assessment chart (appendix 6)
7. Patient safety alert Resources to support safer bowel care for patients at risk of autonomic dysreflexia (appendix 7)

Contact Details

Spinal Nursing team	020 3448 3568/3019/3150
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UCLH Urology uclh.urologyreferralscc@nhs.net
Uclh.urology.queries@nhs.net

11.0 Guidance Implementation

- 11.1 This guideline will be disseminated through the divisional structure for Neurosurgery and Neurology to the relevant Nursing teams and wards.
- 11.2 Education on the use of the guideline should be incorporated into divisional and local training/teaching, with emphasis decided by the needs of the different staff groups.

12.0 Review, Monitoring & Compliance

Key process for which compliance or effectiveness is being monitored	Monitoring method (i.e. audit, report, on-going committee review, survey etc.)	Job title and department of person responsible for leading the monitoring	Frequency of the monitoring activity	Monitoring Committee responsible for receiving the monitoring report/audit results etc.	Committee responsible for ensuring that action plans are completed
Clinical outcome of SCI patients	Audit on SI Survey of stakeholder familiarity Numbers of staff trained	Divisional clinical directors for Queen Square	Annually	Queen Square Clinical Governance Committees	QSD Clinical Governance Committees

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Appendix 1**Acute Spinal Cord Injury (ASCI) Bladder Management**

Insert long-term/silicone indwelling catheter within 3 hours of admission and record hourly urine output (achieve 0.5mls/Kg/hr). Continue until patient is out of spinal shock
(Spinal Shock – the cessation of all reflexive activity below the level of the injury)

Doctor to complete ASIA assessment within 24 hours

The ASIA score will determine whether the patient is a Complete or Incomplete Injury

Incomplete Spinal Cord Injury
 (Preservation of power or sensation in the sacral nerves)

Complete Spinal Cord Injury
 (No power or sensation in the sacral nerves)

Reflexive Bladder
 (Usually T12 and above)

Arreflexive Bladder
 (Usually L1 and below)

TWOC within 2 Weeks
 (Mon –Thurs am only)
 Consider supra pubic
 catheter (SPC) for SCI of C6
 and above
 Consider anticholinergic/alpha blocker

At NHNN Uro-Neuro to be
 informed. Plan to remove the IDC
 and start ISC within 2 weeks.
 For Injuries above C6 - early
 consideration of SPC
 At all other UCH sites refer to
 Urology as early as possible and
 plan for the same regime as
 NHNN

Patient passed urine?

Yes

No

Document pre and
 post bladder scans
 for 48 hours.
 If post void
 volumes are 20%
 or more of pre void
 volume (i.e. pre
 void volume
 500mls, post void
 volume 100mls)
 inform uro-
 neurology. Patient
 will need to perform
 ISC +/-
 anticholinergic/alpa
 blocker medication

At NHNN –
 inform uro-
 Neuro and they
 will teach ISC
 All other UCH
 sites – seek
 advice and
 support from
 urology.
 Consider Supra
 pubic catheter
 for SCI of C6
 and above

Repeat ASIA, when and if symptoms change
 and alter bladder management as appropriate
**Patients can be in spinal shock between 24
 hours and 6 weeks post injury and may
 change to a reflexive pattern**

Mixed Pattern Bladder

Usually from injuries directly at the conus, T12-L2
 These patients may need a reflexive or arreflexive
 pathway or a combination of both
 The appropriate pathway needs to be selected after
 careful assessment and examination of the patient.
 Be aware that the bladder function of these patients
 may fluctuate and change and need ongoing expert
 assessment

Appendix 2

Acute Spinal Cord Injury (ASCI) Bowel Management

Bowel management regime to begin within 24 hours
 Doctor to complete ASIA assessment within 24 hours
 The ASIA assessment will determine which regime the patient will need
 Incomplete Injury = Reflexive bowel regime. Complete Injury = Flaccid regime
 Be aware that spinal shock can last up to 6 weeks post injury

Incomplete Spinal Cord Injury
 Preservation of power or sensation in the sacral nerves

Reflexive Bowel Regime
 (Usually T12 and above)

Daily Glycerine Suppositories 4g x 2
Macrogol 25ml / 1 sachet (Mane) & **Senna** 2 tabs Nocte
 (Review and titrate at 72 hours)

Breakfast incl. hot drink
 Abdominal massage

Insert the suppositories after checking the rectum is clear
 – if faeces are in the rectum, remove digitally and then insert two glycerine suppositories. Wait 20-30 minutes. Place patient in a position to defecate.

Patient passes stool?

Yes

20 minutes after defaecation perform a PR check. If the rectum has faeces in situ, digitally remove and check rectum again 20 minutes later. Repeat this until the rectum is empty when checked.

No

Digital Removal of Faeces
 PR check – apply gentle, clockwise pressure on the internal sphincter with the pad of the index finger. When the internal sphincter opens and faeces move into the rectum, remove digitally. Repeat until the rectum is empty

Complete Spinal Cord Injury
 No power or sensation in the sacral nerves

Flaccid Bowel Regime
 (Usually L1 and below)

Macrogol 25ml / 1 sachet (Mane) & **Senna** 2 tabs Nocte
 (Review and titrate at 72 hours)

Breakfast incl. HOT drink
 Abdominal massage

Digital Removal of Faeces PR check – apply gentle, clockwise pressure on the internal sphincter with the pad of the index finger. When the internal sphincter opens and faeces move into the rectum, remove digitally. Repeat until the rectum is empty

PR check 20 minutes after rectum is emptied. If there is faeces present in the rectum, digitally remove and check the rectum again in 20 minutes. Repeat this until the rectum is empty.

Mixed Pattern

Patients with an injury at the conus (T12-L2) may present with a combination of reflexive and flaccid bowel pattern. Management of these patients is often complex and their regime will need to be titrated to their individual needs. Often they will need to be on a combination of the reflexive and flaccid bowel management regimes. They will need ongoing expert assessment and management.

Appendix 3

BLADDER CARE IN ACUTE SPINAL CORD INJURY (ASCI)

Patients with spinal cord injury (SCI) will be left with a neurogenic bladder, caused by neurological damage to the spinal cord. Bladder dysfunction is broadly divided into Reflexic (spastic) and Areflexic (flaccid) or Mixed (combination of flaccid / spastic). The type and symptoms will depend on the level and whether there is any motor / sensory function preservation of the sacral nerves (complete versus incomplete SCI)

Reflexic (spastic)

Patients with a Neurological Level of Injury (NLI) at T12 and above, are likely to have a reflexic bladder function.

These patients experience involuntary bladder contractions and feel the need to urinate even when there is little or no urine in the bladder. Bladder (detrusor) contractions are typically poorly coordinated with the external sphincter and involuntary contractions occur (Shenot, 2016).

Symptoms include:

Frequency, urgency, nocturia, spastic paralysis with sensory deficits, urine leakage (Shenot, 2016) and sexual dysfunction and erectile dysfunction in men, pelvic pain, bladder over-extension. These symptoms can vary by site and severity of the lesion and fall into two categories:

- **Detrusor Sphincter Dysnergia (DSD)**

A lack of coordination between contraction and relaxation of the detrusor muscle and external sphincter. Both contract at the same time causing high intravesical pressure, incomplete voiding, and risk of vesico-ureteric reflux (Ontario Neurotrauma Foundation and Department of Family Medicine, 2012).

- **Neurogenic Detrusor Overactivity (NDO)**

An increased activity of the detrusor muscle usually combined with impaired contractility and a lack of coordination between the sphincter and the detrusor. The detrusor contracts at small volumes, and against an overactive sphincter (Middleton et al., 2013).

All reflexic patients are at high risk of vesico-urethral reflux (due to the high bladder pressures) which may result in recurrent bladder infections, bladder stones, hydronephrosis, pyelonephritis, and renal failure (Middleton et al., 2013).

Initial Treatment

Immediate, continuous catheterization is needed. If the patient's catheter is bypassing, consideration of the use of an anticholinergic/alpha-blocker drug, as this may ease the spasms preventing urine bypassing.

Bladder training can commence as soon as appropriate by clamping off the catheter and asking the patient to tell you when they feel the need to pass urine. NB Do not leave the catheter clamped for more than 4 hours without draining it.

As soon as possible, patient should be trialed without a catheter (TWOC) (NICE, 2014). If the patient manages to pass urine, ward staff must record 24-48 hours worth of pre and post void bladder scans, along with the volume of urine passed. If post void volumes are 20% or more of the pre void volume, the patient is likely to need intermittent self-catheterization (ISC) training to empty bladder completely.

Note should also be taken if a patient passes small amounts frequently (this denotes that the patient may need an anticholinergic /alpha-blocker drug). They must be seen by uro-neurology (NHNN), urology (Tower) as soon as possible as an inpatient.

Intermittent self-catheterization is preferable to indwelling urethral catheterization, which has a high risk of recurrent UTIs and in men, a high risk of urethritis, periurethritis, prostatic abscesses, and urethral fistulas. Suprapubic catheterization may be used if patients cannot self-catheterize (see appendix A for acute spinal cord bladder management pathway).

If patients fail TWOC, it is ideal for them to be taught ISC straight away. If this is not possible, the re-insertion of a long term indwelling catheter is needed, and early consideration given to a supra pubic catheter.

Areflexic (Flaccid) Bladder

Patients with a cord injury below NLI T12 are likely to have an areflexic (flaccid), bladder function. These patients are likely to experience little or no contraction of the detrusor muscle. This causes urine retention with no sensation of a filling or full bladder (painless retention of urine), until the urine overflows. Bladder volumes are large with a low pressure and contractions are absent (Dorsher and McIntosh, 2012)

Symptoms include:

Urinary retention, overflow incontinence, sexual dysfunction and erectile dysfunction in men, pelvic pain, bladder over-extension, autonomic dysreflexia (Shenot, 2016).

Initial Treatment

Initial treatment consists of immediate, continuous catheterization. As soon as possible, the patient should be trialed without a catheter (NICE, 2014). If the patient manages to pass urine, ward staff must record 24-48 hours' worth of pre and post void bladder scans, along with the volume of urine passed. If post void volumes are 20% or more of the pre void volume, the patient is likely to need intermittent catheterization training to empty bladder completely. They must be seen by uro-neurology (NHNN) / urology (Tower) ASAP as an inpatient.

Intermittent self-catheterization is preferable to indwelling urethral catheterization, which has a high risk of recurrent UTIs and in men, a high risk of urethritis, periurethritis, prostatic abscesses, and urethral fistulas. Suprapubic catheterization may be used if patients cannot self-catheterize (see appendix 1 for acute spinal cord bladder management pathway).

If patients fail TWOC, it is ideal for them to be taught ISC straight away. If this is not possible, the re-insertion of a long term indwelling catheter is needed and early consideration of a supra pubic catheter.

On-going treatment for all SCI Patients

All SCI patients with bladder involvement should be referred on to Uro-Neurology. This can happen as an outpatient. They will need further diagnosis and on-going treatment, which may involve imaging, teaching other manoeuvres, (e.g. Valsalva) cystoscopy, urodynamic testing and possibly surgery (NICE, 2012).

Mixed Pattern

A mixed pattern may occur if the injury is at the conus terminale, (where the spinal cord finishes and the cauda equina begins) around T11-L2 resulting in a flaccid **or** spastic bladder. These patients can experience a combination of symptoms from both reflexic and areflexic bladders.

It is important to note that acutely after cord injury, spinal shock occurs, where all bladders are flaccid generally for up to 2 weeks (but may be longer or much shorter), and after this time as the cord recovers, the more permanent clinical picture will arise.

Additional Notes

Autonomic Dysreflexia (AD) is a potentially dangerous complication for patients with a spinal cord injury (SCI) at T6 or above. In AD, an individual's blood pressure may rise to dangerous levels and if not treated can lead to stroke and possibly death. AD usually occurs when there is a noxious (irritating) stimulus below the level of the injury. Bladder distention and UTI's are frequent causes of AD. Symptoms include headache, tachycardia or bradycardia, facial flush, perspiration, and a stuffy nose (United Spinal Association, 2015).

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Appendix 4

BOWEL CARE IN ACUTE SPINAL CORD INJURY (ASCI)

Patients with Spinal cord injury (SCI) will be left with a neurogenic bowel, caused by neurological damage to the spinal cord. Neurogenic bowel dysfunction denotes impaired sensory and / or motor control of the ano-rectum, resulting in reduced or absent voluntary control of defecation. Stool transit through the bowel, therefore, may be very slow. Patients are at high risk of complications ranging from severe constipation to bowel perforation. The combination of both constipation and impaired continence is termed neurogenic bowel dysfunction (Ash, 2005).

Emmanuel (2010) and Coggrave& Norton (2013), point out the major negative impact poor bowel management has on a patient's quality of life, being the second most distressing aspect of injury after loss of mobility. It inhibits the ability to socialize, live independently, impacts on sexuality and is amongst the most restrictive consequences of their injury. It is imperative that effective evidence based management begins immediately.

A neurogenic bowel describes colon dysfunction due to loss of normal sensory and motor control as a result of neurological damage (Chung and Emmanuel, 2006). The neurological level of injury (NLI) determines neurogenic bowel behavior. This is divided into: Reflexic (spastic), Areflexic (flaccid) and mixed pattern.

Reflexic (spastic)

Patients with a NLI above the conus medullaris (usually T12) are likely to have an upper motor neurone (Reflexic) bowel. Patients have intact reflex arcs from the conus medullaris, characterized by increased colonic wall and anal tones. This allows patients to use the maintained tone to aid movement of faeces through the colon and defaecate. Studies show ([Krassioukov, Janice](#) et al (2010) that, voluntary control of the external anal sphincter is disrupted and the sphincter remains tight, thereby promoting stool retention.

Reflexic bowels are typically associated with constipation and faecal retention; at least in part due to external anal sphincter over activity (dis-coordinated sphincters).

Symptoms include:

Loss / impaired sensory perception of the need to defaecate

Impaired voluntary contraction of anal sphincter

Tone is increased and modulation of peristalsis from the brain is lost causing uncoordinated spasms.

Transit time of faeces through the colon is increased

Constipation / Faecal impaction

Overflow incontinence

Haemorrhoids

Less common - Faecal incontinence (MASCIP, 2012)

Initial Treatment

According to current MASCIP clinical guidelines (2012) a bowel management plan should be instigated within 24 hours of an injury to the cord.

At UCLH it is imperative that our local policy is followed:

Establishing a daily routine at the same time is key to gaining control over continence and avoiding a build-up of stool in the colon. It is best to establish this in the morning when the gastro colic reflex is strongest (This is a physiological response that occurs when one eats. The stretch receptors in the stomach initiate gut motility and peristalsis within the GI tract hence aiding defecation).

Regular oral slow acting laxatives should be given at night (e.g. Sennakot, 2 tablets) and faster acting in the morning with breakfast e.g. Macrogol, to achieve a Bristol stool (Lewis & Heaton, 1997) consistency of 4.

The ward pharmacist should be made aware that the patient is following a regime and the drugs should be reviewed at 72 hours and titrated to each individual's needs.

Patient should be given their breakfast and a **HOT** drink. This is to stimulate the gastrocolic reflex. Patients need to be educated on diet. Generally a diet containing 5 portions of fruit / veg and increased levels of wholegrain should be adhered to. Good breakfast suggestions are Weetabix, bran flakes, brown bread, fruit juice and a hot drink. Patients, from a bowel point of view, should be encouraged to have an increased level of fluid, but bladder management needs to be taken into account here.

Gentle rectal stimulant should be inserted after breakfast (e.g. Glycerine suppository 4g x 2). This helps to trigger evacuation of stool; oral laxatives alone will not achieve this.

The patient or staff member should then undertake abdominal massage. This is firm but gentle massage of the abdomen in a clockwise direction using the heel of ones hand; starting in the bottom right hand corner, in a continuous motion up across and down the abdomen, following the usual lie of the colon. Studies show that abdominal massage produces a measurable response in the rectum and colon (Coggrave et al 2007).

Digital rectal stimulation should occur around 20 minutes after suppositories are inserted, either at the bedside or over the toilet. (See additional notes re RCN guidance). This technique is used to increase reflex muscular activity in the colon to aid stool propulsion.

Once the patient has defecated, digital rectal examination of the rectum must occur to ensure the rectum is empty. If it is not empty the nurse / patient should progress to digital removal of faeces and if required this cycle may need to occur again. (See additional notes re RCN for clarification of all qualified nurses responsibilities (RCN 2012).

Areflexic (Flaccid) Bowel

Patients with a NLI below the conus medullaris, (usually T12 or below) are likely to have an areflexic (flaccid) bowel function. They lose centrally-mediated spinal cord peristalsis; that is no reflex activity is spared in the colon, causing slow stool propulsion. Areflexic bowels are commonly associated with constipation and a significant risk of incontinence due to the atonic (flaccid) external anal sphincter and lack of control over the pelvic floor and levator ani muscle that causes the lumen of the rectum to open (Krassioukov, Janice et al 2010).

Symptoms include:

Loss / impaired sensory perception of the need to defaecate

Nil or impaired voluntary contraction of anal sphincter

No spinal cord peristalsis, loss of effective faecal transit.

Flaccid external anal sphincter, pelvic floor

Transit time of faeces through the colon is considerably increased

Faecal incontinence

Constipation / Faecal impaction

Overflow incontinence

Haemorrhoids (MASCIP, 2012)

Initial Treatment

According to current MASCIP clinical guidelines (2012) a bowel management plan should be instigated within 24 hours.

At UCLH it is imperative that our local policy is followed:

Establishing a daily routine at the same time is key to gaining control over continence and avoiding a build-up of stool in the colon. It is best to establish this in the morning when the gastro colic reflex is strongest (This is a

physiological response to eating: Stretch receptors in the stomach initiate gut motility and peristalsis within the GI tract hence aiding defecation).

Regular oral slow acting laxatives should be given at night (e.g. Sennakot 2 tablets) and faster acting in the morning with breakfast e.g. Macrogol, to achieve a Bristol stool (Lewis & Heaton, 1997) consistency of 2-3.

The ward pharmacist should be made aware that the patient is following a regime and the drugs should be reviewed at 72 hours and titrated to each individual's needs.

Patient should be given their breakfast and a **HOT** drink. This is to stimulate the gastrocolic reflex. Patients need to be educated on diet. A diet containing 5 portions of fruit / veg and increased levels of wholegrain should be adhered to. Patients, from a bowel point of view, should be encouraged to have an increased level of fluid, but bladder management needs to be taken into account here.

The patient or staff member should then undertake abdominal massage. Massage the abdomen, starting in the bottom right hand corner, in a clockwise direction using the heel of the hand. Pressure is applied firmly, but gently, in a continuous motion up across and down the abdomen, following the usual lie of the colon. Studies show that abdominal massage produces a measurable response in the rectum and colon (Coggrave et al 2007)

Digital removal of faeces (DRF). (See additional notes re RCN guidance). This aids in the removal of faeces from the rectum, avoiding distention and impaction. The nurse / patient should continue DRF until the rectum is clear of faeces. (see Flow diagram Appendix 1).

Mixed Pattern

A mixed pattern may occur if the injury is at the conus terminale, (where the spinal cord finishes and the cauda equina begins) around T11-L2 resulting in

a flaccid **or** spastic bowel. These patients can experience a combination of both reflexic and areflexic bowel symptoms. These patients treatment should follow the areflexive pattern initially and the specialist spinal nurses should be contacted via the neurosurgical secretaries for further advice.

For further information or advice on the management of Neurogenic bowel for patients at **Queen Square**, please contact the complex spine neurosurgical unit secretaries on:

020 3448 3568/3150/3395

Additional Notes

It is important to note that, acutely after cord injury, spinal shock occurs where all bowels are flaccid, generally for up to 2 weeks (but may be longer or much shorter). After this time, as the cord recovers, the more permanent clinical picture will arise.

Autonomic Dysreflexia (AD), is a potentially dangerous complication for patients with a spinal cord injury (SCI) at T6 or above. In AD, an individual's blood pressure may rise to dangerous levels and if not treated can lead to stroke and possibly death. AD usually occurs because of a noxious (irritating) stimulus below the level of the injury. Severe constipation / impaction and performing bowel management can occasionally cause AD. Symptoms include headache, tachycardia or bradycardia, facial flush, perspiration, and a stuffy nose (United Spinal Association, 2015).

Faecal impaction

“Symptoms of constipation (either with or without overflow incontinence), in the presence of hard faecal mass obstructing the rectum or colon” (Hussain 2014).

Overflow incontinence - If a patient is constipated or has faecal impaction, the liquid element of stool bypasses the hard impacted stool and leaks out. This is often mistaken for 'diarrhoea' by nurses, all laxatives are stopped: this cessation makes the situation worse. It is therefore imperative that it is understood that overflow incontinence is considered before any changes to medication are made.

Digital removal faeces- If you would like more information on undertaking digital removal faeces (DRF) / digital rectal examination (DRE), please refer to RCN Guidance for nurses:

'Management of lower bowel dysfunction, including DRE and DRF'

<https://www.rcn.org.uk/professional-development/publications/pub-003226>

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Guidelines for Management of Neurogenic Bowel Dysfunction in Individuals with Central Neurological Conditions MASCIIP, Middlesex

National Institute Clinical Excellence (June 2007); Faecal incontinence in adults: management

<https://www.nice.org.uk/Guidance/CG49>

Appendix 5

Bristol Stool Chart

Type 1		Separate hard lumps, like nuts (hard to pass)
Type 2		Sausage-shaped but lumpy
Type 3		Like a sausage but with cracks on its surface
Type 4		Like a sausage or snake, smooth and soft
Type 5		Soft blobs with clear-cut edges (passed easily)
Type 6		Fluffy pieces with ragged edges, a mushy stool
Type 7		Watery, no solid pieces. Entirely Liquid

Patient Name _____ Date/Time of Exam _____

Examiner Name _____ Signature _____

RIGHT

**MOTOR
KEY MUSCLES**

**SENSORY
KEY SENSORY POINTS**
Light Touch (LTR) Pin Prick (PPR)

UER
(Upper Extremity Right)

Elbow flexors **C5**
Wrist extensors **C6**
Elbow extensors **C7**
Finger flexors **C8**
Finger abductors (little finger) **T1**

Comments (Non-key Muscle? Reason for NT? Pain?):

LER
(Lower Extremity Right)

Hip flexors **L2**
Knee extensors **L3**
Ankle dorsiflexors **L4**
Long toe extensors **L5**
Ankle plantar flexors **S1**

(VAC) Voluntary Anal Contraction
(Yes/No) ☐

RIGHT TOTALS
(MAXIMUM) (50) (56) (56)

MOTOR SUBSCORES

UER ☐ + UEL ☐ = **UEMS TOTAL** ☐
MAX (25) (25) (50)

LER ☐ + LEL ☐ = **LEMS TOTAL** ☐
MAX (25) (25) (50)

**NEUROLOGICAL
LEVELS**

Steps 1-5 for classification
as on reverse

1. SENSORY **R** **L**
2. MOTOR

**3. NEUROLOGICAL
LEVEL OF INJURY** ☐
(NLI)

4. COMPLETE OR INCOMPLETE? ☐
Incomplete = Any sensory or motor function in S4-5

5. ASIA IMPAIRMENT SCALE (AIS) ☐

(In complete injuries only)
**ZONE OF PARTIAL
PRESERVATION**
Most caudal level with any innervation

SENSORY **R** **L**
MOTOR

LEFT

**MOTOR
KEY MUSCLES**

**SENSORY
KEY SENSORY POINTS**
Light Touch (LTL) Pin Prick (PPL)

UEL
(Upper Extremity Left)

Elbow flexors **C5**
Wrist extensors **C6**
Elbow extensors **C7**
Finger flexors **C8**
Finger abductors (little finger) **T1**

**MOTOR
(SCORING ON REVERSE SIDE)**

0 = total paralysis
1 = palpable or visible contraction
2 = active movement, gravity eliminated
3 = active movement, against gravity
4 = active movement, against some resistance
5 = active movement, against full resistance
5* = normal corrected for pain/disuse
NT = not testable

**SENSORY
(SCORING ON REVERSE SIDE)**

0 = absent
1 = altered
2 = normal
NT = not testable

LEL
(Lower Extremity Left)

Hip flexors **L2**
Knee extensors **L3**
Ankle dorsiflexors **L4**
Long toe extensors **L5**
Ankle plantar flexors **S1**

(DAP) Deep Anal Pressure
(Yes/No) ☐

LEFT TOTALS
(MAXIMUM) (50) (56) (56)

SENSORY SUBSCORES

LTR ☐ + LTL ☐ = **LT TOTAL** ☐
MAX (56) (56) (112)

PPR ☐ + PPL ☐ = **PP TOTAL** ☐
MAX (56) (56) (112)

Muscle Function Grading

- 0** = total paralysis
1 = palpable or visible contraction
2 = active movement, full range of motion (ROM) with gravity eliminated
3 = active movement, full ROM against gravity
4 = active movement, full ROM against gravity and moderate resistance in a muscle specific position
5 = (normal) active movement, full ROM against gravity and full resistance in a functional muscle position expected from an otherwise unimpaired person
5* = (normal) active movement, full ROM against gravity and sufficient resistance to be considered normal if identified inhibiting factors (i.e. pain, disuse) were not present
NT = not testable (i.e. due to immobilization, severe pain such that the patient cannot be graded, amputation of limb, or contracture of > 50% of the normal ROM)

Sensory Grading

- 0** = Absent
1 = Altered, either decreased/impaired sensation or hypersensitivity
2 = Normal
NT = Not testable

When to Test Non-Key Muscles:

In a patient with an apparent AIS B classification, non-key muscle functions more than 3 levels below the motor level on each side should be tested to most accurately classify the injury (differentiate between AIS B and C).

Movement	Root level
Shoulder: Flexion, extension, abduction, adduction, internal and external rotation Elbow: Supination	C5
Elbow: Pronation Wrist: Flexion	C6
Finger: Flexion at proximal joint, extension. Thumb: Flexion, extension and abduction in plane of thumb	C7
Finger: Flexion at MCP joint Thumb: Opposition, adduction and abduction perpendicular to palm	C8
Finger: Abduction of the index finger	T1
Hip: Adduction	L2
Hip: External rotation	L3
Hip: Extension, abduction, internal rotation Knee: Flexion Ankle: Inversion and eversion Toe: MP and IP extension	L4
Hallux and Toe: DIP and PIP flexion and abduction	L5
Hallux: Adduction	S1

ASIA Impairment Scale (AIS)

A = Complete. No sensory or motor function is preserved in the sacral segments S4-5.

B = Sensory Incomplete. Sensory but not motor function is preserved below the neurological level and includes the sacral segments S4-5 (light touch or pin prick at S4-5 or deep anal pressure) AND no motor function is preserved more than three levels below the motor level on either side of the body.

C = Motor Incomplete. Motor function is preserved at the most caudal sacral segments for voluntary anal contraction (VAC) OR the patient meets the criteria for sensory incomplete status (sensory function preserved at the most caudal sacral segments (S4-S5) by LT, PP or DAP), and has some sparing of motor function more than three levels below the ipsilateral motor level on either side of the body.
 (This includes key or non-key muscle functions to determine motor incomplete status.) For AIS C – less than half of key muscle functions below the single NLI have a muscle grade ≥ 3 .

D = Motor Incomplete. Motor incomplete status as defined above, with at least half (half or more) of key muscle functions below the single NLI having a muscle grade ≥ 3 .

E = Normal. If sensation and motor function as tested with the ISNCSCI are graded as normal in all segments, and the patient had prior deficits, then the AIS grade is E. Someone without an initial SCI does not receive an AIS grade.

Using ND: To document the sensory, motor and NLI levels, the ASIA Impairment Scale grade, and/or the zone of partial preservation (ZPP) when they are unable to be determined based on the examination results.

ASIA
 AMERICAN SPINAL INJURY ASSOCIATION
 INTERNATIONAL STANDARDS FOR NEUROLOGICAL
 CLASSIFICATION OF SPINAL CORD INJURY

ISCOS
 INTERNATIONAL SPINAL CORD SOCIETY

Steps in Classification

The following order is recommended for determining the classification of individuals with SCI.

1. Determine sensory levels for right and left sides.

The sensory level is the most caudal, intact dermatome for both pin prick and light touch sensation.

2. Determine motor levels for right and left sides.

Defined by the lowest key muscle function that has a grade of at least 3 (on supine testing), providing the key muscle functions represented by segments above that level are judged to be intact (graded as a 5).

Note: in regions where there is no myotome to test, the motor level is presumed to be the same as the sensory level, if testable motor function above that level is also normal.

3. Determine the neurological level of injury (NLI)

This refers to the most caudal segment of the cord with intact sensation and antigravity (3 or more) muscle function strength, provided that there is normal (intact) sensory and motor function rostrally respectively.

The NLI is the most cephalad of the sensory and motor levels determined in steps 1 and 2.

4. Determine whether the injury is Complete or Incomplete.

(i.e. absence or presence of sacral sparing)

*If voluntary anal contraction = **No** AND all S4-5 sensory scores = **0** AND deep anal pressure = **No**, then injury is **Complete**.*

*Otherwise, injury is **Incomplete**.*

5. Determine ASIA Impairment Scale (AIS) Grade:

Is injury Complete? If YES, AIS=A and can record ZPP (lowest dermatome or myotome on each side with some preservation)
NO ↓

Is injury Motor Complete? If YES, AIS=B

NO ↓ (No=voluntary anal contraction OR motor function more than three levels below the motor level on a given side, if the patient has sensory incomplete classification)

Are at least half (half or more) of the key muscles below the neurological level of injury graded 3 or better?

NO ↓
AIS=C

YES ↓
AIS=D

If sensation and motor function is normal in all segments, AIS=E

Note: AIS E is used in follow-up testing when an individual with a documented SCI has recovered normal function. If at initial testing no deficits are found, the individual is neurologically intact; the ASIA Impairment Scale does not apply.



Patient Safety Alert

Resources to support safer bowel care for patients at risk of autonomic dysreflexia

25 July 2018

Alert reference number: NHS/PSA/RE/2018/005

Resource Alert

Patients with spinal cord injury or neurological conditions may have neurogenic bowel dysfunction, which often means they depend on routine interventional bowel care, including the digital (manual) removal of faeces (DRF).

Some of these patients, especially those with spinal cord injury above T6, are particularly susceptible to the potentially life-threatening condition autonomic dysreflexia, which is characterised by a rapid rise in blood pressure, risking cerebral haemorrhage and death. A small number of patients who have had a severe stroke or who have severe forms of Parkinson's Disease, multiple sclerosis, cerebral palsy, or spina bifida may also be susceptible to autonomic dysreflexia.

Autonomic dysreflexia can be caused by non-adherence to a patient's usual bowel routine or during or following interventional bowel care. For all of these patients, bowel care is vital for their health and dignity.

Patients have made NHS Improvement aware of difficulties ensuring their regular bowel care is provided when they come into hospital or mental health units, or access other NHS care such as community care. For example, one patient said:

"Despite explaining my situation, I had to wait eight days for an enema. I'd had shoulder surgery so couldn't do it myself. Had the beginnings of autonomic dysreflexia."

A search of the National Reporting and Learning System (NRLS) over a four-and-a-half-year period identified 61 reports of significant delays in providing DRF or an appropriate alternative, including three cases of autonomic dysreflexia. Reports came from acute hospitals, community services and care homes.

The key issues appeared to be a lack of staff with the training and experience to perform DRF (particularly as opportunities to learn and practice are limited outside of spinal injuries units and community teams who regularly undertake this procedure), or an inability to identify staff with the appropriate training.

The search also indicated:

- unclear local policies stating who could perform DRF, including the patient's carers or healthcare professionals from another provider
- lack of knowledge of relevant clinical guidance
- uncertainty over requirement for and provision of training
- uncertainty over using alternative methods of bowel management
- a mistaken belief that this type of care constitutes assault.

Despite a previous Patient Safety Alert,¹ national clinical guidance,^{2,3} profession-specific guidance^{4,5,6,7} and patient resources,⁸ issues persist that cause patients distress and can put them at risk of severe harm or death. Providers have indicated that additional resources to develop and maintain staff skills in this area would be helpful. This alert provides links to a [range of resources](#)⁹ to support safer bowel care for patients at risk of autonomic dysreflexia, and highlights the publication of [NHS England's Excellence in continence care guidance 2018](#)¹⁰ which addresses how providers can overcome implementation challenges.

Actions

Who: All providers of NHS-funded inpatient and community healthcare*

When: To begin as soon as possible and be completed by 25 January 2019

- 1 Identify an appropriate clinical leader to co-ordinate implementation of this alert.
- 2 Using the resources referred to in this alert, review your local clinical policy and guidance relating to bowel assessment and management.
- 3 Review your local education and training provision for interventional bowel management, and develop an action plan to ensure patients have adequate and timely access to staff who are appropriately trained to carry out these procedures, including in the evening and at weekends.
- 4 Share your reviewed local guidance, advice on how to identify staff who can provide DRF, and the key messages in this alert with medical, nursing and other relevant clinical staff.

*GP's would not be expected to lead on policy/training actions but should be aware of the need to act quickly, and appropriately escalate care for affected patients.

Sharing resources and examples of work

If you are aware of any resources or examples of work developed in relation to this alert that you think would be useful to others, please share them with us by emailing patientsafety.enquiries@nhs.net

Alert reference number: NHS/PSA/RE/2018/005

Resource Alert

NRLS search dates and terms

The NRLS was searched for incidents occurring on or after 13 May 2013 if reported by 19 May 2016, and this search was updated for incidents occurring on or after 13 May 2013 if reported between 19 May 2016 and 1 December 2017, using the search terms [%stool% or %faeces% or %bowel%] AND [%manual% or %digital%] AND [%remov% or %evac%]. This combined search yielded 61 relevant reports describing difficulties providing appropriate bowel care, including three incidents where autonomic dysreflexia had occurred.

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2. National Institute for Health and Care Excellence clinical guideline 2014 [CG49] Faecal incontinence in adults: management <https://www.nice.org.uk/guidance/cg49>
3. National Institute for Health and Care Excellence quality standard 2014 [QS 54] Faecal incontinence in adults <https://www.nice.org.uk/guidance/qs54>
4. British Association of Spinal Cord Injury Specialists (BASCIS), Multidisciplinary Association of Spinal Cord Injury Professionals (MASCIP), Spinal Injuries Association (SIA) 2014 Statement on Autonomic Dysreflexia <https://www.spinal.co.uk/wp-content/uploads/2018/06/Statement-on-Autonomic-Dysreflexia-2017.pdf>
5. Royal College of Nursing 2012 Management of lower bowel dysfunction, including digital rectal examination and digital removal of faeces. <https://www.rcn.org.uk/professional-development/publications/pub-003226>
6. Nursing and Midwifery Council 2018. Future nurse: Standards of proficiency for registered nurses <https://www.nmc.org.uk/globalassets/sitedocuments/education-standards/future-nurse-proficiencies.pdf>
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8. Spinal Injury Association 2013 Patient and professional resources <https://www.spinal.co.uk/wp-content/uploads/2017/05/Autonomic-Dysreflexia.pdf>
9. NHS Improvement 2018, Resources to support safer bowel care for patients at risk of autonomic dysreflexia webpage <https://improvement.nhs.uk/resources/resources-to-support-safer-bowel-care-for-patients-at-risk-of-autonomic-dysreflexia>
10. NHS England (updated 2018) Excellence in continence care framework <https://www.england.nhs.uk/publication/excellence-in-continence-care/>

Resources

- The resources to support the implementation of this Alert are available on the NHS Improvement website <https://improvement.nhs.uk/resources/resources-to-support-safer-bowel-care-for-patients-at-risk-of-autonomic-dysreflexia>

Stakeholder engagement

- Royal College of Nursing Continence Forum
- NHS England Excellence in Continence Care Committee
- The Spinal Injuries Association
- National Patient Safety Response Advisory Panel (for a list of members and organisations represented on the panel, see improvement.nhs.uk/resources/patient-safety-alerts/)

Advice for Central Alerting System officers and risk managers

This alert is particularly relevant to all healthcare services where patients with existing spinal cord injury may need care for other conditions, and is likely to include community nursing services and mental health inpatient units, as well as acute hospitals for adults and children, and care homes.

This alert requires central co-ordination of a revision to local policies and training, so should not be widely circulated until these have been developed.

If you are unsure who to direct this alert to, your local continence nurse specialist will be able to signpost the medical, nursing and allied health professional staff and educational leads who will coordinate local implementation of this alert.